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International Journal of Robust and Nonlinear Control

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SPECIAL ISSUE Advances in Quantum Control, Robust Control and Negative Imaginary Systems

RESEARCH ARTICLE OPEN ACCESS

Marginally Strictly Negative Imaginary Systems

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Received: 29 April 2026 | **Revised:** 7 August 2026 | **Accepted:** 21 September 2026

Keywords: marginally strictly negative imaginary systems | negative imaginary systems | resonant control | robust control

ABSTRACT

This article revisits a fundamental question in negative imaginary (NI) systems theory: what is the weakest viable notion of strictness that still preserves the classical feedback stability picture? To address this question, we introduce the notion of marginally strictly negative imaginary (MSNI) systems to accommodate simple nonzero imaginary-axis poles while preserving strictly negative imaginary behavior away from those poles. An MSNI system is defined as the parallel interconnection of a lossless NI system and a strictly negative imaginary system. The proposed notion provides a less restrictive strictness framework suited for resonant control within NI systems theory. A new stability test is then developed for the positive feedback interconnection of a stable NI system and an MSNI system. The proposed test is stated in terms of the well-known loop DC-gain condition together with easily verifiable pointwise assumptions at the imaginary-axis pole locations. A numerical example illustrates the validity of the condition and its effectiveness under resonant tracking requirements.

1 | Introduction

Negative imaginary (NI) systems theory was pioneered by Lanzon and Petersen [1] in 2008. The work initiated a powerful framework in robust control for feedback stability analysis of NI systems that arise in lightly damped mechanical structures and flexible systems with collocated force actuators and position sensors [2–4]. After nearly two decades of development, NI systems theory has gradually matured into a broad research area. It has been extended to infinite-dimensional systems [5], nonlinear systems [6–9], discrete-time systems [10, 11], lossless NI systems [12–15], networked control settings [16, 17], and related generalized feedback-stability frameworks [18], to name a few. At the same time, the original application-oriented perspective remains important. This is particularly relevant in systems with resonant behavior, which is a prominent feature of many practical systems [19]. NI systems theory is valuable in this setting because it connects such frequency-domain characteristics to simple closed-loop stability conditions.

This article is dedicated to Ian Petersen in celebration of his 70th birthday, building on the pioneering results in [1]. The main result of [1] states that the positive feedback interconnection of a stable NI system and a strictly negative imaginary (SNI) system is stable if and only if a loop DC-gain condition is satisfied. This remains one of the most elegant features of NI systems theory, and related extensions continue to characterize feedback stability through simple frequency-domain conditions in broader settings [20, 21]. Classical NI feedback stability results typically require at least one of the systems in the positive feedback interconnection to be SNI [1, 4, 12, 20–22]. The SNI property, defined for stable systems only, enforces strictness in the frequency-domain inequality on the frequency interval $(0, \infty)$. In many feedback control designs involving NI plants, such NI strictness is typically provided by the designed controllers [2, 20–23]. However, the stability requirement under SNI excludes simple nonzero imaginary-axis poles in the controllers. This may be restrictive when resonant poles are introduced deliberately for control purposes.

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One representative example is the proportional resonant controller, which has been widely used in power-electronic systems and high-accuracy mechanical systems [24–29]. In these applications, nonzero imaginary-axis poles are introduced so that prescribed frequencies are embedded in the controller and precise tracking or disturbance rejection can be achieved at those frequencies. As a result, the SNI controllers are somewhat inapplicable in such settings. This naturally leads to the following question: Can the DC-gain stability test be retained while allowing simple nonzero resonant poles on the side that provides NI strictness? Equivalently, the problem is to identify the weakest viable strictness notion in the NI setting that preserves the DC-gain stability picture.

Two existing research directions are particularly relevant to this question. One line of research studies alternative NI strictness notions, including strongly strictly negative imaginary systems, very strictly negative imaginary systems, input strictly negative imaginary systems, output strictly negative imaginary systems, and related variants [30–34]. These notions broaden the available strictness formulations beyond the classical SNI setting, but most of them are formulated for stable systems. Some variants accommodate imaginary-axis poles in special cases, for example, through the notion of time-domain input–output negative imaginary systems [34]. However, they do not directly address how prescribed nonzero imaginary-axis poles may be embedded on the NI strictness-providing side and they do not establish a corresponding feedback stability result for such a setting.

Another line of research studies positive feedback interconnections in which both systems are NI. The feedback stability result in this setting is developed under additional technical assumptions [35]. These assumptions increase the verification burden and move the stability test away from a purely DC-gain condition. More broadly, existing NI feedback-stability results that allow nonzero imaginary-axis poles often require stronger pointwise assumptions at the corresponding pole locations of the other system [32, 35, 36]. Such requirements can be conservative and may exclude some stable feedback interconnections. The discussion later in the article shows that these requirements can be relaxed without losing a stability condition that depends only on the DC gain.

To answer the question raised above, in this article we introduce the notion of marginally strictly negative imaginary (MSNI) systems, inspired in part by the early notion of marginally strictly positive real systems [37] and the recent notions of systems phase [38–40]. An MSNI system is defined as the parallel interconnection of a lossless NI system and an SNI system. The lossless NI part carries the nonzero imaginary-axis poles needed for resonant controller design, while the SNI part preserves SNI behavior away from those poles. Such a structure can be interpreted from the internal model principle [41] introduced half a century ago. The principle indicates that zero steady-state error in reference tracking or disturbance rejection at a prescribed frequency requires the controller to contain the corresponding frequency information. The proposed MSNI decomposition is consistent with the principle and well suited for resonant tracking and feedback stability analysis of NI systems.

The main contributions of this article are as follows. First, the class of MSNI systems is introduced, and its basic properties are stated. Second, a stability theorem is established for the positive feedback interconnection of a stable NI system and an MSNI system, while preserving the DC-gain condition and requiring only a simple pointwise nonsingularity check at the imaginary-axis pole locations of the MSNI system. Third, a physical example is given to illustrate both the stability result and the resulting resonant tracking behavior. A preliminary version of this work has been accepted for presentation at a conference [42]. The present article contains complete proofs and additional technical developments and contributions.

The remainder of the article is organized as follows: Section 2 presents the notation and preliminary definitions. Section 3 introduces the MSNI notion and its basic properties. Section 4 develops the supporting arguments and proves the main stability theorem. Section 5 presents an illustrative example, and Section 6 concludes the article.

2 | Notation and Preliminaries

This section introduces basic notation and recalls several definitions of NI systems needed for the article. It also recalls the feedback system setting and the stability theorem [1] for the positive feedback interconnection of an NI system and an SNI system, which will be extended in the MSNI setting.

2.1 | Notation

Let $\mathbb{C}$ denote the complex plane, and $\mathbb{C}_{>0}$ and $\mathbb{C}_{\geq 0}$ denote the open and closed right-half planes respectively. Let $\mathbb{R}^{n\times n}$ and $\mathbb{C}^{n\times n}$ denote the sets of $n\times n$ real and complex matrices respectively. Let $\mathcal{R}^{n\times n}$ denote the set of $n\times n$, real, rational, proper transfer function matrices, and $\mathcal{RH}_\infty^{n\times n}$ denote the subset of stable transfer function matrices in $\mathcal{R}^{n\times n}$, that is, those with no poles in $\mathbb{C}_{\geq 0}$. The identity matrix of compatible dimension is denoted by I. For a matrix A, the transpose and complex conjugate transpose are denoted by $A^\top$ and A^* respectively. If all eigenvalues of A are real, then $\lambda_{\max}(A)$ denotes the largest eigenvalue of A. A matrix $A\in\mathbb{C}^{n\times n}$ is Hermitian if $A=A^*$, and a matrix $A\in\mathbb{R}^{n\times n}$ is symmetric if $A=A^\top$. For a Hermitian or real symmetric matrix A, $A\geq 0$ (resp. $A>0$) means that A is positive semidefinite (resp. positive definite).

2.2 | Negative Imaginary Systems

Background on NI, SNI, and lossless NI systems can be found in [1, 12, 15, 22]. Only the definitions needed for the development of the MSNI framework are given below.

Throughout this article, we focus on *nonzero* resonant poles on the imaginary axis only and do *not* consider poles at the origin for linear systems for technical simplicity. For a system $G\in\mathcal{R}^{n\times n}$, we define the set of positive imaginary-axis pole frequencies of G by

$$\Omega_G=\{\omega\in(0,\infty):\ s=j\omega \text{ is a pole of } G(s)\}.$$

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Definition 1 ([22]). A system $G \in \mathcal{R}^{n\times n}$ is said to be *negative imaginary (NI)* if

1. G has no poles at the origin and in $\mathbb{C}_{>0}$;
2. $j[G(j\omega) - G(j\omega)^*] \geq 0$ for all $\omega \in (0,\infty) \setminus \Omega_G$;
3. for each $\omega_0 \in \Omega_G$, the pole at $s = j\omega_0$ is simple and the residue matrix $\lim_{s\to j\omega_0}(s - j\omega_0)\, jG(s)$ is Hermitian and positive semidefinite.

Lossless NI systems form a special subclass of NI systems for which the frequency-domain inequality in Definition 1 holds with equality for all $\omega \in (0,\infty) \setminus \Omega_G$.

Definition 2 ([12]). A system $G \in \mathcal{R}^{n\times n}$ is said to be *lossless negative imaginary (lossless NI)* if G is NI and for all $\omega \in (0,\infty) \setminus \Omega_G$,

$$j[G(j\omega) - G(j\omega)^*] = 0.$$

SNI systems form a stable subclass of NI systems and are characterized by a strict frequency-domain inequality on the frequency interval $(0,\infty)$.

Definition 3 ([1]). A system $G \in \mathcal{RH}_\infty^{n\times n}$ is said to be *strictly negative imaginary (SNI)* if for all $\omega \in (0,\infty)$,

$$j[G(j\omega) - G(j\omega)^*] > 0.$$

2.3 | Positive Feedback Interconnections and NI Stability Results

We next present positive feedback interconnections and an existing NI feedback stability theorem.

Consider the positive feedback interconnection shown in Figure 1, where $P \in \mathcal{R}^{n\times n}$ and $C \in \mathcal{R}^{n\times n}$. The external signals are w_1 and w_2, and the internal signals are u_1, u_2, y_1 and y_2. The feedback interconnection is denoted by $[P, C]$ and is described by

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} I & C \\ P & I \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}.$$

Definition 4 ([43, Lemma 5.3]). A positive feedback interconnection of P and C is said to be *stable* if

$$\begin{bmatrix} I & C \\ P & I \end{bmatrix}^{-1} = \begin{bmatrix} I + C(I-PC)^{-1}P & -C(I-PC)^{-1} \\ -(I-PC)^{-1}P & (I-PC)^{-1} \end{bmatrix} \in \mathcal{RH}_\infty^{2n\times 2n}.$$

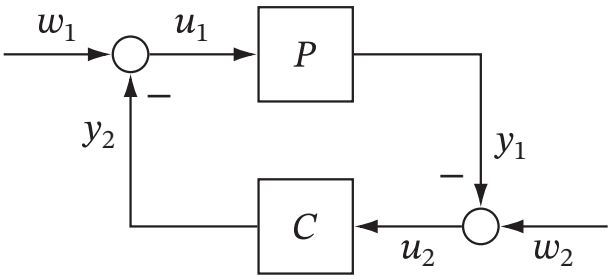


FIGURE 1 | A positive feedback interconnection of $P, C \in \mathcal{R}^{n\times n}$.

The following classical NI stability theorem for the positive feedback interconnection is given in [1, 21].

Theorem 1 ([1, 21]). *Let $P \in \mathcal{R}^{n\times n}$ be NI and $C \in \mathcal{RH}_\infty^{n\times n}$ be SNI. Suppose that at least one of them is strictly proper. Then, the positive feedback interconnection $[P, C]$ is stable if and only if*

$$\lambda_{\max}(P(0)C(0)) < 1. \tag{1}$$

The result yields a simple DC-gain test (1) provided that one of the two systems is SNI. The SNI assumption rules out simple nonzero imaginary-axis poles on that side. We argue that this restriction can be relaxed by introducing the MSNI notion.

3 | MSNI Systems

This section introduces the class of MSNI systems. We begin with a motivating example that highlights the need to accommodate resonant poles and motivates the decomposition used in the MSNI definition.

3.1 | A Motivating Example

In Example 1, the system P is a representative NI example considered in the literature [2, 4, 33, 34], and the system C contains an SNI part $1/(s+1)$. Since both P and the SNI part of C are stable, the setting, that is, the feedback interconnection of $P(s)$ and $1/(s+1)$, remains aligned with Theorem 1. However, the SNI part alone is not sufficient to reflect the resonant control action often introduced at a prescribed nonzero frequency in applications. Such resonant action is relevant in tracking and disturbance rejection problems involving periodic signals at known frequencies [24–29, 36]. An explicit resonant part $1/(s^2+1)$ is included in C to capture this feature. The new term introduces a pair of simple imaginary-axis poles and shapes the closed-loop behavior near the selected frequency. Its use is consistent with the internal model principle, which requires the prescribed frequency to be embedded in the controller [41].

Example 1. Consider

$$P(s) = \frac{2s^2 + s + 1}{(2s+1)(s+1)(s^2+2s+5)},$$
$$C(s) = \frac{1}{s^2+1} + \frac{1}{s+1}.$$

All the poles of P lie in the open left-half plane. Moreover, P is NI. The positive-frequency Nyquist plot in Figure 2 illustrates this property and shows that $P(j)$ lies on the real axis. Therefore, the frequency-domain inequality in the definition of NI systems is not strict at $\omega = 1$ rad/s. For C, the first term $1/(s^2+1)$ is lossless NI and has simple poles at $s = \pm j$. The second term $1/(s+1)$ is clearly SNI. Since $P(j) = 0.1$, the poles of $C(s)$ at $s = \pm j$ are not cancelled in the loop gain $P(s)C(s)$. A numerical root check shows that all zeros of $1 - P(s)C(s)$ lie in the open left-half plane. Hence the positive feedback interconnection is stable by [43, Theorem 5.7]. Clearly, such a conclusion *cannot* be drawn from Theorem 1.

Note that $P(0)C(0) = 0.4$ and both systems are strictly proper. Together with $P(j) \neq 0$, this indicates that the DC-gain test (1)

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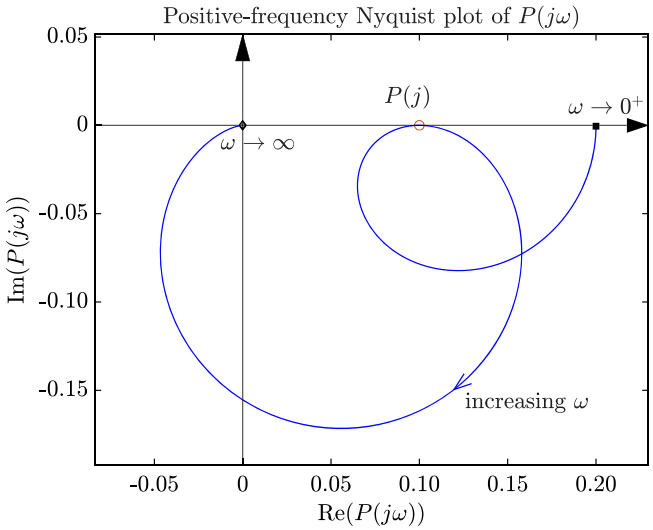


FIGURE 2 | The positive-frequency Nyquist plot of $P(j\omega)$, where $P(s) = \frac{2s^2+s+1}{(2s+1)(s+1)(s^2+2s+5)}$ is given in Example 1.

may be sufficient when the resonant poles are not cancelled in $P(s)C(s)$.

3.2 | MSNI Definition

We now introduce the MSNI definition to formalize the idea described in Example 1.

Definition 5 (MSNI system). A system $G \in \mathcal{R}^{n\times n}$ is said to be *marginally strictly negative imaginary (MSNI)* if it admits the following decomposition

$$G = G_{\mathrm{m}} + G_{\mathrm{s}},$$

where G_{m} is lossless NI, $G_{\mathrm{m}}(\infty) = G(\infty)$, and G_{s} is SNI.

Note that the decomposition in Definition 5 is unique since the feedthrough term of G is assigned to G_{m}, and thereby G_{s} is always strictly proper. The following remark shows that SNI behavior is retained away from the imaginary-axis poles.

Remark 1. Since $G_{\mathrm{s}} \in \mathcal{RH}_\infty^{n\times n}$, it has no poles on the imaginary axis. Hence $\Omega_{G_{\mathrm{s}}} = \emptyset$ and $\Omega_G = \Omega_{G_{\mathrm{m}}}$. Since G_{m} is lossless NI, for every $\omega \in (0,\infty) \setminus \Omega_G$,

$$j[G_{\mathrm{m}}(j\omega) - G_{\mathrm{m}}(j\omega)^*] = 0,$$

and therefore

$$j[G(j\omega) - G(j\omega)^*] = j[G_{\mathrm{s}}(j\omega) - G_{\mathrm{s}}(j\omega)^*] > 0.$$

The SNI term G_{s} is essential for enforcing this SNI behavior away from the resonant poles.

For instance, a simple MSNI system can be written in the form

$$G(s) = G_{\mathrm{m}}(s) + G_{\mathrm{s}}(s),$$

where

$$G_{\mathrm{m}}(s) = \frac{k_{\mathrm{m}}}{s^2+\omega_0^2}, \quad G_{\mathrm{s}}(s) = \frac{k_{\mathrm{s}}}{s+a},$$

with $k_{\mathrm{m}} > 0$, $k_{\mathrm{s}} > 0$, $\omega_0 > 0$, and $a > 0$. The first term is lossless NI and carries the poles at $s = \pm j\omega_0$, and the second term is SNI. Hence G is MSNI by Definition 5. The system $C(s)$ in Example 1 corresponds to the choice $k_{\mathrm{m}} = k_{\mathrm{s}} = 1$, $\omega_0 = 1$, and $a = 1$.

4 | Main Results on Feedback Stability

This section extends the classical NI stability result by relaxing the SNI system to an MSNI system with simple nonzero imaginary-axis poles under the restriction that the NI system is stable. This relaxation comes at the cost of an additional requirement to exclude unstable pole-zero cancellation in the closed right-half plane, as discussed in Example 1. That is, the unstable poles of the loop gain $P(s)C(s)$, counted with multiplicity, should equal the sum of the unstable poles of $P(s)$ and $C(s)$. In the MIMO setting, this is enforced by additional assumptions on the stable NI system. These assumptions are full normal rank [43, Section 3.11] together with the absence of transmission zeros at the imaginary-axis pole locations of the MSNI system, so that pole-zero cancellation is ruled out at those imaginary-axis pole locations. The stability condition can still be stated as the same DC-gain test, as elaborated in the following theorem:

Theorem 2. *Let $P \in \mathcal{RH}_\infty^{n\times n}$ be NI with full normal rank and $C \in \mathcal{R}^{n\times n}$ be MSNI. Suppose that at least one of them is strictly proper. Suppose that $P(s)$ has no transmission zeros at any point $s = j\omega$ where $C(s)$ has a pole. Then, the positive feedback interconnection $[P, C]$ is stable if and only if*

$$\lambda_{\max}(P(0)C(0)) < 1.$$

Proof. Denote by

$$\Omega = \{\omega \in (0,\infty) : s = j\omega \text{ is a pole of } C(s)\}.$$

The proof has three steps. Step 1 excludes unstable pole-zero cancellation between P and C in $\mathbb{C}_{\geq 0}$. This step is specific to the present MSNI setting, because C may have simple nonzero imaginary-axis poles. Step 2 excludes transmission zeros of $I - P(s)C(s)$ on the positive imaginary axis. The argument follows an idea similar to that used in the proofs of [21, Lemma 6] and [33, Lemma 18], but the pole case requires additional local analysis. Once these two steps are established, Step 3 follows the state-space argument in the proof of [1, Theorem 5], with the exclusion on the imaginary axis supplied here by Step 1 and Step 2.

Step 1: We exclude unstable pole-zero cancellations between P and C in $\mathbb{C}_{\geq 0}$.

Since $P \in \mathcal{RH}_\infty^{n\times n}$ and $C \in \mathcal{R}^{n\times n}$ is MSNI, the only possible pole-zero cancellations in the closed right-half plane occur at imaginary-axis poles of $C(s)$, that is, at $s = j\omega$ for $\omega \in \Omega$. By assumption, $P(s)$ has full normal rank and no transmission zeros at $s = j\omega$ for all $\omega \in \Omega$. Therefore, none of the poles of $C(s)$ at $s = j\omega$ such that $\omega \in \Omega$ can be cancelled by zeros of $P(s)$. Thus, there are no pole-zero cancellations in the closed right-half plane.

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Step 2: We show that $I - P(s)C(s)$ has no transmission zeros on the $j\omega$-axis for any $\omega \in (0, \infty)$. Two separate cases are considered below, namely (a) $\omega \in (0, \infty) \setminus \Omega$ and (b) $\omega \in \Omega$.

Case (a): $\omega_0 \in (0, \infty) \setminus \Omega$. This is the regular case where neither P nor C has a pole at $s = j\omega_0$.

Let $\omega_0 \in (0, \infty) \setminus \Omega$ be arbitrary. By the definition of NI and Remark 1,

$$jP(j\omega_0) + (jP(j\omega_0))^* \geq 0,$$
$$jC(j\omega_0) + (jC(j\omega_0))^* > 0.$$

It follows that $\det(jC(j\omega_0)) \neq 0$ and

$$(jC(j\omega_0))^{-1} + (jC(j\omega_0))^{-*} > 0.$$

Hence,

$$[(jC(j\omega_0))^{-1} + jP(j\omega_0)] + [(jC(j\omega_0))^{-1} + jP(j\omega_0)]^* > 0.$$

This indicates that $(jC(j\omega_0))^{-1} + jP(j\omega_0)$ is nonsingular. Consequently,

$$I - P(j\omega_0)C(j\omega_0) = [(jC(j\omega_0))^{-1} + jP(j\omega_0)]jC(j\omega_0)$$

is nonsingular. Therefore, $I - P(s)C(s)$ has no transmission zeros at $s = j\omega_0$ for all $\omega_0 \in (0, \infty) \setminus \Omega$.

Case (b): $\omega_0 \in \Omega$. This is the pole case, where the MSNI decomposition of C and the pointwise assumptions on P are used.

Let $\omega_0 \in \Omega$ be arbitrary. Since C is MSNI, Definition 5 implies $C = C_m + C_s$, where C_m is lossless NI and C_s is SNI. Hence,

$$jC_s(j\omega_0) + (jC_s(j\omega_0))^* > 0.$$

Since $P \in \mathcal{RH}_\infty^{n\times n}$ has full normal rank and no transmission zero at $s = j\omega_0$, it follows from [43, Corollary 3.30] that $P(j\omega_0)$ is nonsingular. Since P is NI and $P(j\omega_0)$ is nonsingular, we have

$$(jP(j\omega_0))^{-1} + (jP(j\omega_0))^{-*} \geq 0.$$

Therefore,

$$[jC_s(j\omega_0) + (jP(j\omega_0))^{-1}] + [jC_s(j\omega_0) + (jP(j\omega_0))^{-1}]^* > 0. \quad (2)$$

There exists $\delta > 0$ such that $P(s)$ is analytic and nonsingular and $C(s)$ is analytic for all s satisfying $0 < |s - j\omega_0| < \delta$, because $P \in \mathcal{RH}_\infty^{n\times n}$ with $P(j\omega_0)$ being nonsingular and $s = j\omega_0$ being an isolated pole of $C(s)$. Fix such a sufficiently small δ. For all real ω such that $s = j\omega$ and $0 < |s - j\omega_0| < \delta$, Remark 1 gives

$$jC(j\omega) + (jC(j\omega))^* = jC_s(j\omega) + (jC_s(j\omega))^* > 0.$$

Therefore,

$$[jC(j\omega) + (jP(j\omega))^{-1}] + [jC(j\omega) + (jP(j\omega))^{-1}]^* > 0,$$

which implies that $jC(j\omega) + (jP(j\omega))^{-1}$ is nonsingular for all real ω with $0 < |\omega - \omega_0| < \delta$. Hence, $jC(s) + (jP(s))^{-1}$ has full normal rank.

For all s satisfying $0 < |s - j\omega_0| < \delta$, the identity

$$I - P(s)C(s) = (jP(s))(jC(s) + (jP(s))^{-1})$$

holds. Since $jP(s)$ is analytic and nonsingular on $\{s : |s - j\omega_0| < \delta\}$, left multiplication by $jP(s)$ does not create or remove a transmission zero at $s = j\omega_0$. Hence, $I - P(s)C(s)$ has a transmission zero if and only if $jC(s) + (jP(s))^{-1}$ has a transmission zero at $s = j\omega_0$.

It remains to rule out a transmission zero of $jC(s) + (jP(s))^{-1}$ at $s = j\omega_0$. Assume that such a transmission zero exists. Since $jC(s) + (jP(s))^{-1}$ has full normal rank, the transmission zero definition in [44, Section 27.4] implies that there exists a rational vector $u(s)$ such that $u(j\omega_0)$ is finite and nonzero, and

$$\lim_{s\to j\omega_0} [jC(s) + (jP(s))^{-1}]u(s) = 0.$$

Restricting the limit to the imaginary axis $s = j\omega$ yields

$$\lim_{\omega\to\omega_0} [jC(j\omega) + (jP(j\omega))^{-1}]u(j\omega) = 0.$$

Premultiplying by $u(j\omega)^*$ and taking the Hermitian part gives

$$\begin{aligned}
&\lim_{\omega\to\omega_0} u(j\omega)^*([jC(j\omega) + (jP(j\omega))^{-1}] \\
&\qquad + [jC(j\omega) + (jP(j\omega))^{-1}]^*)u(j\omega) \\
&= u(j\omega_0)^*([jC_s(j\omega_0) + (jP(j\omega_0))^{-1}] \\
&\qquad + [jC_s(j\omega_0) + (jP(j\omega_0))^{-1}]^*)u(j\omega_0) \\
&= 0.
\end{aligned}$$

However, this contradicts (2). Therefore, $I - P(s)C(s)$ has no transmission zeros at $s = j\omega_0$ for any $\omega_0 \in \Omega$.

Combining these two cases, $I - P(s)C(s)$ has no transmission zeros on the $j\omega$ axis for all $\omega \in (0, \infty)$.

Step 3: Once Step 1 and Step 2 are established, the remainder of the proof follows the state-space argument in [1, Theorem 5]. We highlight only the parts needed for the present extension.

Let (A_1, B_1, C_1, D_1) and (A_2, B_2, C_2, D_2) be minimal state-space realizations of P and C, respectively. Since P is stable NI, we can apply [1, Lemma 1] to P. Moreover, since every MSNI system is NI and has no poles at the origin, we can apply [22, Lemma 7] to C. It follows that A_1 is Hurwitz, $\det(A_2) \neq 0$, $D_1 = D_1^\top$, $D_2 = D_2^\top$, and there exist real symmetric matrices $Y_1 = Y_1^\top > 0$ and $Y_2 = Y_2^\top > 0$ satisfying the corresponding NI lemma conditions.

Then, the same matrices Φ and T as in the proof of [1, Theorem 5] are introduced. In particular,

$$T = \begin{bmatrix} Y_1^{-1} - C_1^\top D_2 C_1 & -C_1^\top C_2 \\ -C_2^\top C_1 & Y_2^{-1} - C_2^\top D_1 C_2 \end{bmatrix}.$$

Let $\hat{A}$ denote the state matrix of the corresponding realization of $(I - PC)^{-1}$. Then, following the same construction as in the proof of [1, Theorem 5], we obtain

$$\hat{A} = \Phi T, \quad \Phi + \Phi^\top \leq 0.$$

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Since at least one of P and C is strictly proper, it follows that $I - P(\infty)C(\infty)$ is nonsingular. By Step 1, there is no pole-zero cancellation between P and C in $\mathbb{C}_{\geq 0}$. Hence, by [43, Theorem 5.7], the positive feedback interconnection $[P, C]$ is stable in the sense of Definition 4 if and only if

$$(I - PC)^{-1} \in \mathcal{RH}_\infty^{n\times n}.$$

Moreover, the corresponding state-space realization is stabilizable and detectable. Hence, $(I - PC)^{-1} \in \mathcal{RH}_\infty^{n\times n}$ holds if and only if $\hat{A}$ is Hurwitz.

The above relations imply

$$T\hat{A} + \hat{A}^\top T \leq 0.$$

In the proof of [1, Theorem 5], this inequality is used together with the exclusion of transmission zeros of $I - P(s)C(s)$ on the imaginary axis away from the origin to establish the equivalence between $\hat{A}$ being Hurwitz and $T > 0$. Here this exclusion is provided by Step 2. Hence, the same argument shows that $\hat{A}$ is Hurwitz if and only if $T > 0$.

It remains to convert $T > 0$ to the DC-gain condition. Consider first the case where P is strictly proper. Then $D_1 = 0$, and

$$T = \begin{bmatrix} Y_1^{-1} - C_1^\top D_2 C_1 & -C_1^\top C_2 \\ -C_2^\top C_1 & Y_2^{-1} \end{bmatrix}.$$

Since $Y_2 > 0$, the Schur complement gives

$$T > 0 \;\Leftrightarrow\; Y_1^{-1} - C_1^\top (D_2 + C_2 Y_2 C_2^\top) C_1 > 0.$$

Using $P(0) = C_1 Y_1 C_1^\top$ and $C(0) = D_2 + C_2 Y_2 C_2^\top$, a straightforward matrix manipulation shows that

$$Y_1^{-1} - C_1^\top (D_2 + C_2 Y_2 C_2^\top) C_1 > 0 \;\Leftrightarrow\; \lambda_{\max}(P(0)C(0)) < 1.$$

If instead C is strictly proper, the same conclusion follows after interchanging the roles of P and C. If both P and C are strictly proper, then $D_1 = D_2 = 0$, so the above calculation reduces to the same conclusion. Thus, the above discussion shows that $T > 0$ is equivalent to $\lambda_{\max}(P(0)C(0)) < 1$ in each case.

Therefore, under the assumptions of the theorem, $[P, C]$ is stable if and only if $\lambda_{\max}(P(0)C(0)) < 1$. □

Theorem 2 shows that the SNI assumption in Theorem 1 may be stronger than what is actually needed for the frequency-domain stability conclusion developed here, since it excludes nonzero imaginary-axis poles altogether. Such poles do not by themselves make closed-loop stability impossible. In the NI setting considered here, poles at the origin are excluded for the sake of simplicity, and the conditions at 0 and ∞ are already handled under the strict properness assumption together with the DC-gain condition. What matters at the remaining nonzero imaginary-axis pole locations is whether they give rise to pole-zero cancellation between P and C. Hence, the role of the additional assumption is confined to these nonzero imaginary-axis pole locations.

In the SISO case, the additional assumption in Theorem 2 reduces to $P(j\omega) \neq 0$ for every imaginary-axis pole $j\omega$ of C. The stability condition then becomes the scalar test $P(0)C(0) < 1$ together with finitely many pointwise nonzero checks. Example 1 is a direct instance of this SISO case. In the general MIMO setting, Theorem 2 preserves the DC-gain condition in Theorem 1. The additional assumptions are that $P \in \mathcal{RH}_\infty^{n\times n}$ has full normal rank and no transmission zeros at the imaginary-axis pole locations of C. These conditions ensure that $P(j\omega)$ is nonsingular at each such location; see [43, Lemma 3.29 and Corollary 3.30]. Under this pointwise nonsingularity, the stability condition in Theorem 2 can still be stated as the DC-gain test. Note that the no-transmission-zero assumption is sufficient, but may not be necessary. This deserves further research.

Compared with Theorem 2, the results in [20, 21] assume an SNI system in the positive feedback interconnection and thus rule out imaginary-axis poles on that side. A possible alternative is to relocate the lossless NI part from the MSNI side to the stable NI side and then apply [21]. However, this requires additional assumptions and a separate NI verification step for the transformed system, as in [45]. The MSNI framework avoids this relocation and yields a direct and easily verifiable stability test for the original feedback interconnection. In addition, related results often require pointwise NI strictness on the system at the imaginary-axis pole locations of the other system [32, 35, 36]. In our MSNI setting, SNI behavior is retained by the SNI part in C, so no pointwise SNI assumption is required on the stable NI system at those pole locations. This feature is illustrated further in the next section through a physical example.

5 | Illustrative Example

This section provides a physical illustration of the stability result in Section 4. It considers a collocated mechanical model that is classical in NI systems theory. The example shows that the plant is stable NI, and it does not satisfy the strict frequency-domain inequality at the resonant frequency. Nevertheless, Theorem 2 still reduces the stability test to a DC-gain condition together with a pointwise nonsingularity check at the controller pole.

5.1 | Mechanical Plant Model

A two-mass spring–damper system with collocated force inputs and displacement outputs is used, following [1, 22]. The plant is shown in Figure 3. Let $q(t) = \begin{bmatrix} q_1(t) & q_2(t) \end{bmatrix}^\top$ be the displacement output vector and $f(t) = \begin{bmatrix} f_1(t) & f_2(t) \end{bmatrix}^\top$ be the collocated force input vector. In Figure 1, the plant port signals are identified as

$$u_1(t) = f(t),$$
$$y_1(t) = q(t).$$

The equation of motion is

$$M\ddot{q}(t) + D\dot{q}(t) + Kq(t) = f(t),$$

where

$$M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad K = \begin{bmatrix} 3 & -2 \\ -2 & 3 \end{bmatrix}.$$

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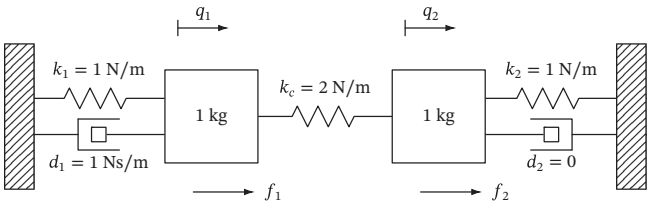


FIGURE 3 | A two-mass spring-damper plant with collocated force inputs f_1, f_2 and displacement outputs q_1, q_2.

The transfer function matrix from f to q is

$$P(s) = (Ms^2 + Ds + K)^{-1} = \frac{\begin{bmatrix} s^2+3 & 2 \\ 2 & s^2+s+3 \end{bmatrix}}{s^4 + s^3 + 6s^2 + 3s + 5}.$$

The plant is stable since all poles of P lie in the open left-half plane. Moreover, it is NI. For any $\omega > 0$, $s = j\omega$ is neither a pole nor a transmission zero of $P(s)$. Then, it follows from [43, Corollary 3.30] that $P(j\omega)$ is nonsingular, and hence

$$j[P(j\omega) - P(j\omega)^*] = 2\omega P(j\omega) D P(j\omega)^* \geq 0.$$

The frequency-domain inequality is not strict because D is singular. In fact, D has rank one, and therefore $j[P(j\omega) - P(j\omega)^*]$ is rank deficient for all $\omega > 0$. In particular, no pointwise NI strictness is available at the resonant pole location $s = j\omega_0$, where $\omega_0 = 1$ rad/s, used in the controller below.

5.2 | MSNI Controller

The controller is chosen in the MSNI form

$$C(s) = C_{\mathrm{m}}(s) + C_{\mathrm{s}}(s)$$

with

$$C_{\mathrm{m}}(s) = \begin{bmatrix} \dfrac{k_{\mathrm{m}}}{s^2 + \omega_0^2} & 0 \\ 0 & 0 \end{bmatrix}, \quad C_{\mathrm{s}}(s) = \frac{k_{\mathrm{s}}}{s + a} I,$$

where $k_{\mathrm{m}} > 0$, $k_{\mathrm{s}} > 0$, $\omega_0 > 0$, and $a > 0$.

Here C_{m} is lossless NI and carries simple poles at $s = \pm j\omega_0$ in the $(1,1)$ entry. Moreover, $C_{\mathrm{s}} \in \mathcal{RH}_\infty^{2\times 2}$ and satisfies

$$j[C_{\mathrm{s}}(j\omega) - C_{\mathrm{s}}(j\omega)^*] = \frac{2k_{\mathrm{s}}\omega}{\omega^2 + a^2} I > 0 \quad \forall \omega > 0.$$

Hence C_{s} is SNI. Therefore, C is MSNI by Definition 5.

Let $k_{\mathrm{m}} = k_{\mathrm{s}} = 0.5$, $\omega_0 = 1$, and $a = 1$. The controller has imaginary-axis poles only at $s = \pm j$. A direct evaluation gives $\det(P(j)) = -0.5j$, so the pointwise nonsingularity requirement holds. The DC-gain test is also satisfied, with $\lambda_{\max}(P(0)C(0)) \approx 0.770$. Hence, the positive feedback interconnection $[P, C]$ is stable by Theorem 2. This example lies outside the scope of Theorem 1, since C is not SNI. Moreover, it is also not covered by existing NI stability results, because P does not satisfy pointwise NI strictness at the pole location of C [32–36]. Even so, Theorem 2 still applies and yields the stability conclusion directly. We next examine the closed-loop behavior near the resonant frequency.

5.3 | Resonant Tracking on a Selected Channel

For the positive feedback interconnection shown in Figure 1, we focus on the first componentwise channel from $[w_2]_1$ to $[y_1]_1$. To illustrate resonant tracking on this channel, we set

$$w_2(t) = \begin{bmatrix} \sin(\omega_0 t) & 0 \end{bmatrix}^\top.$$

The closed-loop transfer function matrix from w_2 to y_1 is

$$T(s) = -(I - P(s)C(s))^{-1} P(s)C(s).$$

Hence the transfer function from $[w_2]_1$ to $[y_1]_1$ is the $(1,1)$ entry of $T(s)$, namely $T_{11}(s)$. To analyze why tracking occurs on this channel, we consider the corresponding error path. Since $u_2 = w_2 - y_1$, the error transfer function matrix from w_2 to u_2 is

$$S(s) = I - T(s) = (I - P(s)C(s))^{-1},$$

whose $(1,1)$ entry is denoted by $S_{11}(s)$. Thus, $[u_2]_1 = S_{11}(s)[w_2]_1$.

This is consistent with the internal model principle [41]. For the chosen input, $[w_2]_1 = \omega_0/(s^2 + \omega_0^2)$. The resonant term $C_{\mathrm{m}}(s)$ introduced in Section 5.2 has poles at $s = \pm j\omega_0$ in the first control channel and embeds the same resonant mode. The nonsingularity of $P(\pm j\omega_0)$ prevents these controller poles from being cancelled in the loop. For the present plant and controller structure, the selected-channel error transfer function $S_{11}(s)$ has corresponding zero factors at $s = \pm j\omega_0$, so the poles of $[w_2]_1$ at those points are cancelled in $S_{11}(s)[w_2]_1$. Since the positive feedback interconnection is stable by Theorem 2, the remaining factor is stable. Therefore $[w_2(t)]_1 - [y_1(t)]_1 \to 0$ as $t \to \infty$.

This conclusion is also reflected in the frequency response of $T_{11}(s)$. As shown in Figure 4, $|T_{11}(j\omega)| \approx 1$ and $\angle T_{11}(j\omega) \approx 0$ for

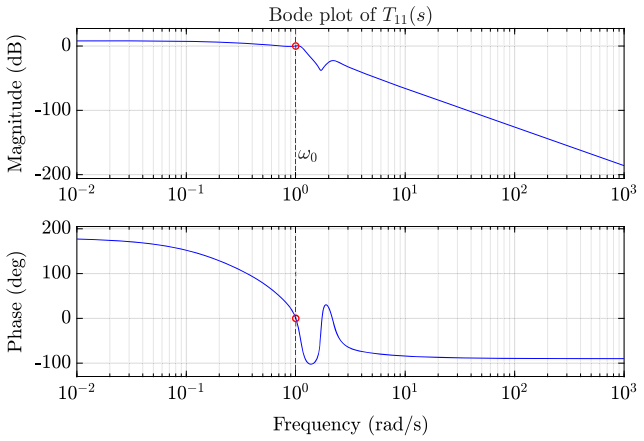


FIGURE 4 | Bode plot of $T_{11}(s)$, the closed-loop transfer function from $[w_2]_1$ to $[y_1]_1$, with $\omega_0 = 1$ rad/s.

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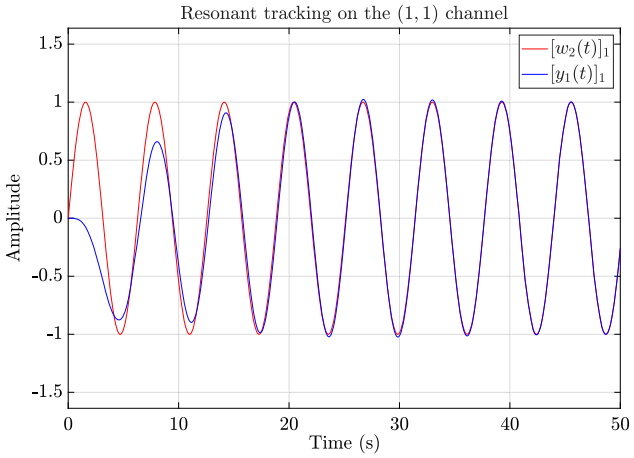


FIGURE 5 | Time response of $[y_1(t)]_1$ to $[w_2(t)]_1 = \sin(\omega_0 t)$ with $\omega_0 = 1$ rad/s.

all ω in a neighborhood of ω_0. Figure 5 shows the corresponding response $[y_1(t)]_1$. In steady state, $[y_1(t)]_1$ tracks $[w_2(t)]_1 = \sin(\omega_0 t)$ closely.

6 | Conclusion

This article introduces marginally strictly negative imaginary systems as the weakest strictness notion in the present setting. This notion permits simple nonzero imaginary-axis poles and is therefore suited to resonant control. Based on the MSNI framework, a DC-gain stability test is developed for the positive feedback interconnection of a stable NI system and an MSNI system, together with an easily verifiable pointwise nonsingularity check. A physical example is included to illustrate the resulting resonant tracking behavior and to demonstrate how the proposed framework can be used in controller design.

Acknowledgments

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Funding

This work was supported by the Engineering and Physical Sciences Research Council [grant number UKRI2416] and the National Science and Technology Council of Taiwan [grant numbers: 113-2222-E-110-002-MY3, 114-2622-8-110-003, 115-2218-E-007-003 and 115-2221-E-110-058-MY2].

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

References

1. A. Lanzon and I. R. Petersen, "Stability Robustness of a Feedback Interconnection of Systems With Negative Imaginary Frequency Response," *IEEE Transactions on Automatic Control* 53, no. 4 (2008): 1042–1046.

2. I. R. Petersen and A. Lanzon, "Feedback Control of Negative-Imaginary Systems," *IEEE Control Systems Magazine* 30, no. 5 (2010): 54–72.

3. S. K. Das, H. R. Pota, and I. R. Petersen, "Resonant Controller Design for a Piezoelectric Tube Scanner: A Mixed Negative-Imaginary and Small-Gain Approach," *IEEE Transactions on Control Systems Technology* 22, no. 5 (2014): 1899–1906.

4. I. R. Petersen, "Negative Imaginary Systems Theory and Applications," *Annual Reviews in Control* 42 (2016): 309–318.

5. M. R. Opmeer, "Infinite-Dimensional Negative Imaginary Systems," *IEEE Transactions on Automatic Control* 56, no. 12 (2011): 2973–2976.

6. D. Zhao, C. Chen, and S. Z. Khong, "A Frequency-Domain Approach to Nonlinear Negative Imaginary Systems Analysis," *Automatica* 146 (2022): 110604.

7. K. Shi, I. R. Petersen, and I. G. Vladimirov, "Output Feedback Consensus for Networked Heterogeneous Nonlinear Negative-Imaginary Systems With Free-Body Motion," *IEEE Transactions on Automatic Control* 68, no. 9 (2023): 5536–5543.

8. A. G. Ghallab, M. A. Mabrok, and I. R. Petersen, "Negative Imaginary Systems Theory for Nonlinear Systems: A Dissipativity Approach," *IEEE Transactions on Automatic Control* 70, no. 12 (2025): 8120–8132.

9. C. Chen, Q. Sun, S. Z. Khong, and A. Lanzon, "A Nonlinear Negative Imaginary Theorem for Feedback Stability," in *Proceedings of the 65th IEEE Conference on Decision and Control* (IEEE, 2026) To Appear.

10. A. Ferrante, A. Lanzon, and L. Ntogramatzidis, "Discrete-Time Negative Imaginary Systems," *Automatica* 79 (2017): 1–10.

11. M. Liu and J. Xiong, "Properties and Stability Analysis of Discrete-Time Negative Imaginary Systems," *Automatica* 83 (2017): 58–64.

12. J. Xiong, I. R. Petersen, and A. Lanzon, "On Lossless Negative Imaginary Systems," *Automatica* 48, no. 6 (2012): 1213–1217.

13. S. Rao and P. Rapisarda, "An Algebraic Approach to the Realization of Lossless Negative Imaginary Behaviors," *SIAM Journal on Control and Optimization* 50, no. 3 (2012): 1700–1720.

14. M. Liu and G. Chen, "Partial-Fraction Expansion of Lossless Negative Imaginary Property and a Generalized Lossless Negative Imaginary Lemma," in *Proceedings of the 57th IEEE Conference on Decision and Control* (IEEE, 2018), 7089–7094.

15. M. Liu, X. Jing, and G. Chen, "Necessary and Sufficient Conditions for Lossless Negative Imaginary Systems," *Journal of the Franklin Institute* 357, no. 4 (2020): 2330–2353.

16. J. Wang, A. Lanzon, and I. R. Petersen, "Robust Output Feedback Consensus for Networked Negative-Imaginary Systems," *IEEE Transactions on Automatic Control* 60, no. 9 (2015): 2547–2552.

17. J. Wang, A. Lanzon, and I. R. Petersen, "Robust Cooperative Control of Multiple Heterogeneous Negative-Imaginary Systems," *Automatica* 61 (2015): 64–72.

18. S. Z. Khong, "Feedback Stability of Generalized Positive Real and Negative Imaginary Systems," *IEEE Transactions on Automatic Control* 68, no. 10 (2023): 6285–6290.

19. L. Meirovitch, *Dynamics and Control of Structures* (John Wiley & Sons, 1990).

20. M. A. Mabrok, A. G. Kallapur, I. R. Petersen, and A. Lanzon, "Generalizing Negative Imaginary Systems Theory to Include Free Body Dynamics: Control of Highly Resonant Structures With Free Body Motion," *IEEE Transactions on Automatic Control* 59, no. 10 (2014): 2692–2707.

21. A. Lanzon and H. J. Chen, "Feedback Stability of Negative Imaginary Systems," *IEEE Transactions on Automatic Control* 62, no. 11 (2017): 5620–5633.

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22. J. Xiong, I. R. Petersen, and A. Lanzon, "A Negative Imaginary Lemma and the Stability of Interconnections of Linear Negative Imaginary Systems," *IEEE Transactions on Automatic Control* 55, no. 10 (2010): 2342–2347.

23. S. Z. Khong, D. Zhao, and A. Lanzon, "Converse Negative Imaginary Theorems," *Automatica* 165 (2024): 111682.

24. R. Teodorescu, F. Blaabjerg, M. Liserre, and P. C. Loh, "Proportional-Resonant Controllers and Filters for Grid-Connected Voltage-Source Converters," *IEE Proceedings-Electric Power Applications* 153, no. 5 (2006): 750–762.

25. S. Fukuda and T. Yoda, "A Novel Current-Tracking Method for Active Filters Based on a Sinusoidal Internal Model," *IEEE Transactions on Industry Applications* 37, no. 3 (2001): 888–895.

26. M. Liserre, R. Teodorescu, and F. Blaabjerg, "Multiple Harmonics Control for Three-Phase Grid Converter Systems With the Use of PI-RES Current Controller in a Rotating Frame," *IEEE Transactions on Power Electronics* 21, no. 3 (2006): 836–841.

27. A. Timbus, M. Liserre, R. Teodorescu, P. Rodriguez, and F. Blaabjerg, "Evaluation of Current Controllers for Distributed Power Generation Systems," *IEEE Transactions on Power Electronics* 24, no. 3 (2009): 654–664.

28. Y. Tao, Z. Zhu, Q. Xu, H. X. Li, and L. Zhu, "Tracking Control of Nanopositioning Stages Using Parallel Resonant Controllers for High-Speed Nonraster Sequential Scanning," *IEEE Transactions on Automation Science and Engineering* 18, no. 3 (2021): 1218–1228.

29. L. Li, Y. Meng, X. Wang, S. S. Aphale, and L. Zhu, "A Proportional Resonant Controller With Switching Scheme for Cross-Coupling Compensation in AFM Sinusoidal Scanning," *IEEE Transactions on Instrumentation and Measurement* 73 (2024): 4510311.

30. P. Bhowmick and S. Patra, "On LTI Output Strictly Negative-Imaginary Systems," *Systems & Control Letters* 100 (2017): 32–42.

31. A. Ferrante, A. Lanzon, and L. Ntogramatzidis, "Foundations of Not Necessarily Rational Negative Imaginary Systems Theory: Relations Between Classes of Negative Imaginary and Positive Real Systems," *IEEE Transactions on Automatic Control* 61, no. 10 (2016): 3052–3057.

32. S. A. Devi, P. Bhowmick, and A. Lanzon, "On State-Space Characterisation for Output Negative Imaginary Systems With Possible Poles at the Origin and Their Internal Stability Result," in *Proceedings of the 62nd IEEE Conference on Decision and Control* (IEEE, 2023), 6547–6552.

33. A. Lanzon and P. Bhowmick, "Characterization of Input-Output Negative Imaginary Systems in a Dissipative Framework," *IEEE Transactions on Automatic Control* 68, no. 2 (2023): 959–974.

34. P. Bhowmick and A. Lanzon, "Dynamic Dissipative Characterisation of Time-Domain Input–Output Negative Imaginary Systems," *Automatica* 164 (2024): 111620.

35. P. Bhowmick and A. Dey, "Negative Imaginary Stability Result Allowing Purely Imaginary Poles in Both the Interconnected Systems," *IEEE Control Systems Letters* 6 (2021): 403–408.

36. S. Z. Khong, S. Hara, and A. Lanzon, "Negative Imaginariness Indices in Feedback Stability Analysis," *IEEE Transactions on Automatic Control* 70, no. 12 (2025): 8321–8328.

37. S. M. Joshi and S. Gupta, "On a Class of Marginally Stable Positive-Real Systems," *IEEE Transactions on Automatic Control* 41, no. 1 (1996): 152–155.

38. C. Chen, W. Chen, D. Zhao, J. Chen, and L. Qiu, "A Cyclic Small Phase Theorem," *IEEE Transactions on Automatic Control* 71, no. 3 (2026): 1676–1691.

39. W. Chen, D. Wang, S. Z. Khong, and L. Qiu, "A Phase Theory of Multi-Input Multi-Output Linear Time-Invariant Systems," *SIAM Journal on Control and Optimization* 62, no. 2 (2024): 1235–1260.

40. J. Chen, W. Chen, C. Chen, and L. Qiu, "Phase Preservation of N-Port Networks Under General Connections," *IEEE Transactions on Automatic Control* 70, no. 4 (2025): 2346–2361.

41. B. A. Francis and W. M. Wonham, "The Internal Model Principle of Control Theory," *Automatica* 12, no. 5 (1976): 457–465.

42. T. Zhang, C. Chen, S. Z. Khong, and A. Lanzon, "On the Strictness of Negative Imaginariness for Linear Systems," in *Proceedings of the 65th IEEE Conference on Decision and Control* (IEEE, 2026) To Appear.

43. K. Zhou, J. C. Doyle, and K. Glover, *Robust and Optimal Control* (Prentice-Hall, 1996).

44. M. Dahleh, M. A. Dahleh, and G. Verghese, "Lectures on Dynamic Systems and Control, Chapter 27: Poles and Zeros of MIMO Systems," MIT OpenCourseWare, Massachusetts Institute of Technology, Course 6.241J/16.338J (Spring 2011), (2011).

45. Y. H. Su, P. Bhowmick, and A. Lanzon, "Properties of Interconnected Negative Imaginary Systems and Extension to Formation-Containment Control of Networked Multi-UAV Systems With Experimental Validation Results," *Asian Journal of Control* 27 (2025): 99–116.

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